Danzel Yap

Diamond Bar, CA 91765
(408) 335-3615
dyap123@gmail.com

SUMMARY

Project Engineer with eight years of building software, robots, and construction projects, now managing structural foundation pours on the $2 billion LA Convention Center expansion while leading the development of AI construction tools.

EXPERIENCE

Webcor Craft, Project Engineer — Los Angeles, CA
January 2026 – Present

- Lead concrete pours for structural foundations and slab on metal deck: inspections, field takeoffs, QC, concrete procurement, temperature sensor installation, team alignment, information flow, and pour execution.
- Detail rebar and produce embed shop drawings (400+ unique embed types), excavation plans, and soffit plans in Revit; fix 3D model issues after thorough QC and own any field corrections traced to my drawings.
- Manage RFIs, submittals, quantity takeoffs, budgets, and look-ahead scheduling with trade partners, superintendents, and the design team.
- Built and implemented AI tools for the steel, MEP, and concrete teams that track 21,000+ pieces of steel, 390+ pieces of electrical equipment, 21,300+ CY of concrete, and 700+ tools across multiple projects.
- Spearhead meetings with internal leadership, IT, Data, and Cybersecurity teams to design a system that scales company-wide.
- Led the evaluation of an external AI vendor and recommended against adoption, saving $75,000.

Clune Construction, Project Management Intern — San Francisco, CA
June 2024 – August 2024

- Processed submittals and wrote RFIs on the Winston & Strawn SF office interior remodel.
- Performed estimating and bid leveling on the Skadden Palo Alto ground-up office and the US Fertility center in Menlo Park.

Cal Poly Pomona Formula E, Nuts & Bolts Lead — Pomona, CA
September 2022 – June 2024

- Led design of a double wishbone suspension and other mechanical elements for a competition electric vehicle.

K–12 STEM Center USA, STEM Instructor — Claremont, CA
October 2022 – March 2026

- Taught robotics, Arduino, coding, and soldering to K–12 students; built and deployed an introductory AI course.

PROJECTS

Integrating AI in Construction
2024 – Present

- Built and deployed a full-stack AI application on a $2 billion project for quantity tracking, embed QC, and steel receiving, plus automations (concrete break tracker, drawing index, steel delivery pipeline) that notify teams when new data arrives.
- Walked active projects including Durango, Resorts World, and Gentle Monster in Las Vegas and the Olympic and Hill tower in Los Angeles, and consulted with executive leadership at 30+ general contractors on AI.
- Managed a $10,000 budget leading a team across logistics, outreach, and software development.

Provectus AI — Consultation
2026 – Present

- Advised contractors with $2B to $6B in revenue on AI strategy at scale: labor efficiency, the inventory procurement pipeline, and immediate ROI.

FIRST Robotics Competition — California
November 2018 – June 2021

- Designed and built a robot arm on a multidisciplinary team; secured grants.

SKILLS

Project Controls, Cost Control, Budgets, RFIs, Submittals, Quantity Takeoff, Shop Drawings, Rebar Detailing, Model QC, Inspections, Look-Ahead Scheduling, Trade Partner Coordination, Quality Control, Field Communication.

SOFTWARE

Autodesk Revit, Revizto, Bluebeam Revu, Autodesk Construction Cloud, Microsoft Office, Google Workspace; Cursor, Claude, Codex; Python, JavaScript, React, Firebase, LLM integration (Anthropic, OpenAI, Gemini); SolidWorks, Autodesk Fusion 360.

EDUCATION

California State Polytechnic University, Pomona
B.S. Mechanical Engineering
Graduated Fall 2025

LANGUAGES

Fluent in Mandarin & English; conversational Cantonese.

INTERESTS

Robotics, applied AI, building (software & hardware), photography & drone videography.